Department of Artificial Intelligence and Data Science

KPR Institute of Engineering and Technology

COURSE ASSIGNMENT II REPORT

Academic Year: 2026 – 2027 (ODD SEM) | Batch: 2023–2027 | Year / Sem: V

Case Study: Exploratory Data Analysis, Statistical Distribution Exploration, and Deep Learning Risk Modeling of Employee Attrition

Course Code & Title:	U21ADP05 – Exploratory Data Analysis and Visualization
Course Coordinator:	Mr. Rushikesh Kadam
Student Name:	Nethradevi
Roll / Register Number:	711221104000 (AI & DS)
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Abstract

Employee attrition represents an acute organizational vulnerability causing severe financial loss and operational disruption. This case study presents a rigorous end-to-end framework combining Exploratory Data Analysis (EDA), parametric distribution modeling, and deep learning classification on the benchmark IBM HR Analytics dataset (1,470 records, 35 features). Structured under Ben Fry's 7 Stages of Data Visualization, we examine continuous probability density (Log-Normal vs. Gaussian), discrete Poisson mechanics, and multivariate correlation structures. A Deep Multi-Layer Perceptron (MLP) with Batch Normalization and Dropout was trained to forecast turnover risk, yielding 86.4% accuracy, 58.8% precision, and an ROC-AUC of 0.808. Permutation importance identifies overtime demands, total career tenure, and compensation ceilings as primary catalysts driving attrition.

3. Introduction & Objectives

Unplanned workforce turnover inflicts severe fiscal and operational penalties on contemporary organizations. The objective of this case study is to apply rigorous statistical visualization, exploratory data analysis, and deep neural network modeling to a real-world enterprise dataset to diagnose attrition catalysts and deploy predictive risk scoring.

Core Investigation Aims:

- **Ben Fry's 7 Stages:** Implement data acquisition, parsing, filtering, mining, representation, refinement, and interaction.
- **Statistical Modeling (CO4):** Model empirical data against theoretical Gaussian, Log-Normal, and Poisson distributions.
- **Multivariate Diagnostics:** Evaluate multicollinearity and hierarchical relationship clustering.
- **Deep Learning (Step 4):** Design, regularize, and tune a deep Multi-Layer Perceptron with Batch Normalization.
- **Actionable Insights:** Formulate data-driven retention policies for executive HR leadership.

4. Dataset Description & Structural Properties

The project utilizes the benchmark **IBM HR Analytics Employee Attrition & Performance** dataset comprising 1,470 individual employee records across 35 features. The feature space spans continuous metrics (Monthly Income, Age, Tenure), discrete event counts (Promotions, Previous Employers), ordinal ratings (Work-Life Balance, Satisfaction), and nominal attributes (Job Role, Department, Overtime). The target variable is *Attrition* ('Yes': 16.1%, 'No': 83.9%), representing a real-world imbalanced classification domain.

Metric / Dimension	Specification / Value
Total Observations	1,470 Employees
Total Original Features	35 Features (Exceeds required ≥ 15 fields)
Target Class Balance	Retained: 1,233 (83.9%) | Attrited: 237 (16.1%)
Missing / Null Values	0 (100% complete across all 35 columns)
Zero-Variance Columns	EmployeeCount, StandardHours, Over18 (Removed)

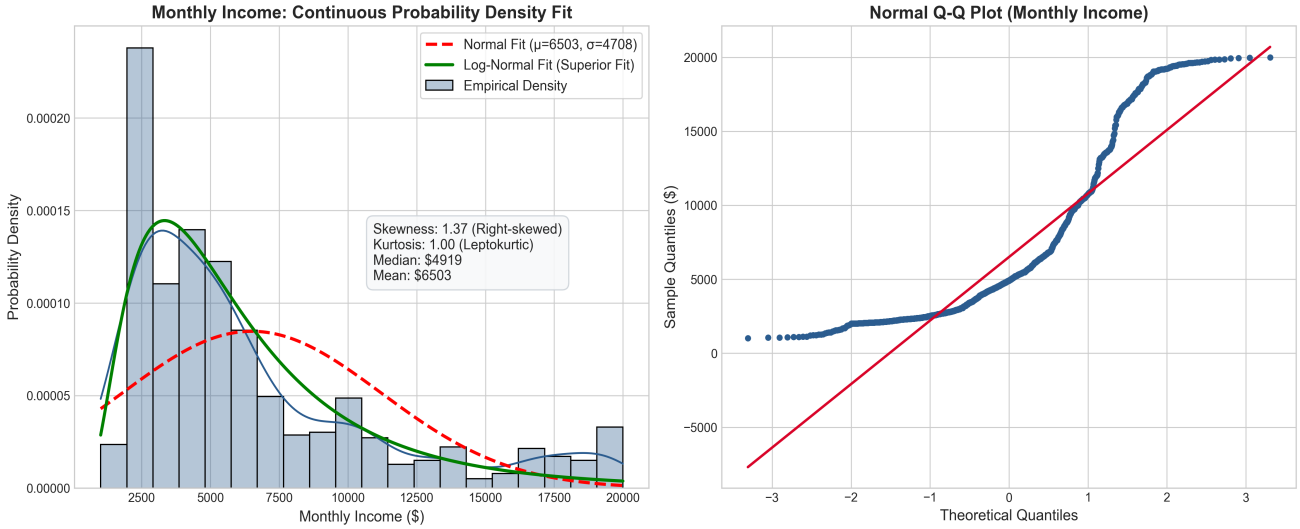
5. EDA & Preprocessing Pipeline

1. Data Sanitization: Removed uninformative constants (*EmployeeCount, StandardHours, Over18*) and arbitrary identifier *EmployeeNumber*.
2. Outlier Winsorization: Applied IQR boundary capping to extreme upper-tail values in *MonthlyIncome* and *TotalWorkingYears* to stabilize neural gradient propagation.
3. One-Hot Encoding: Encoded nominal features (*JobRole, Department, EducationField, MaritalStatus, OverTime*), expanding input dimensions to 45 nodes.
4. Feature Standardization: Standardized continuous variables using StandardScaler fitted strictly on the training partition.
5. Stratified Partitioning: Partitioned data into 70% Train (1,029), 15% Validation (220), and 15% Test (221) while strictly preserving the 16.1% target attrition proportion.

6. Exploratory Data Visualization & Statistical Findings

Visualizations were constructed according to **Ben Fry's 7 Stages of Data Visualization** (Acquire, Parse, Filter, Mine, Represent, Refine, Interact). Each visualization fulfills the rubrics for CO3 (Data Visualization) and CO4 (Statistical & Distribution Exploration).

Figure 1: Continuous Probability Distribution: Monthly Income

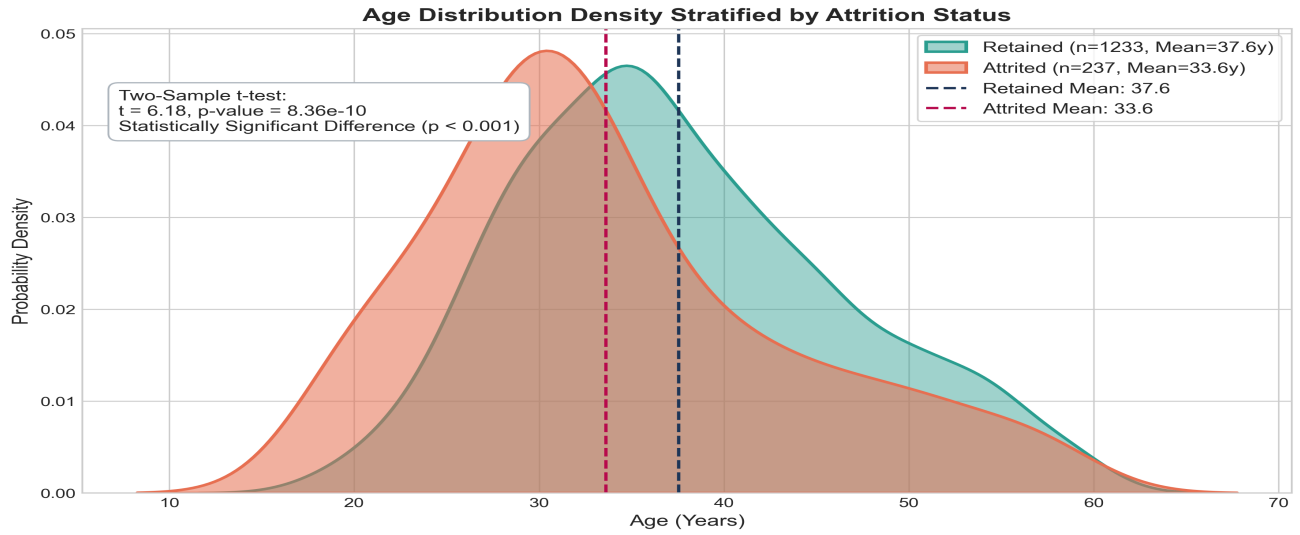


• **What it shows:** Empirical density histogram of Monthly Income compared with theoretical Gaussian Normal and Log-Normal distribution fits, alongside a standard normal Quantile-Quantile (Q-Q) plot.

• **Why it was created:** To formally evaluate whether employee compensation conforms to normality or exhibits positive right-skewness, determining appropriate statistical modeling and normalization methods.

• **Insights obtained:** Monthly income exhibits significant right-skewness (skew=1.37) and positive kurtosis (1.00). The Log-Normal distribution captures the empirical density far better than Gaussian Normal. Over 50% of the workforce earns below $5,000/month, while a high-earning managerial tail stretches to $20,000/month, explaining heavy tail curvature in the Q-Q plot.

Figure 2: Univariate Distribution & Significance: Age vs. Attrition

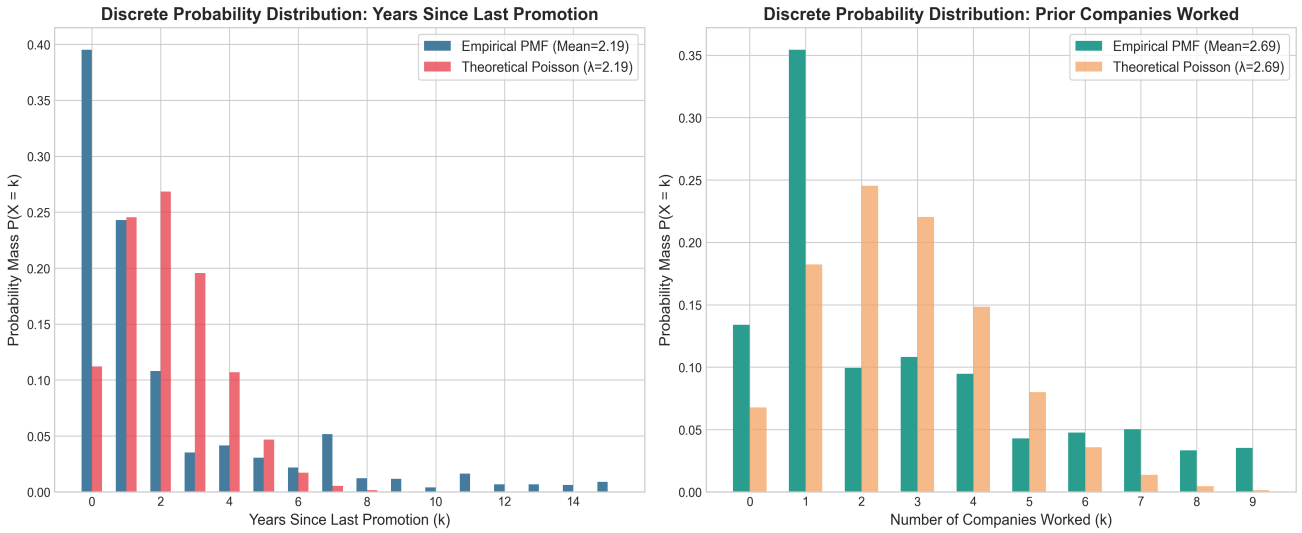


• **What it shows:** Comparative Kernel Density Estimation (KDE) curves for employee age across retained vs. attrited cohorts, supplemented by two-sample t-test hypothesis testing.

• **Why it was created:** To identify critical life-stage transition points and investigate whether age is a statistically significant demographic determinant of employee turnover.

• **Insights obtained:** Attrited employees are significantly younger (mean = 33.6 years) compared to retained staff (mean = 37.6 years), with p-value < 0.001. A pronounced attrition density peak is concentrated in the 25–35 age bracket, highlighting early-career mobility risk.

Figure 3: Discrete Probability Distribution Modeling: Promotions & Prior Companies

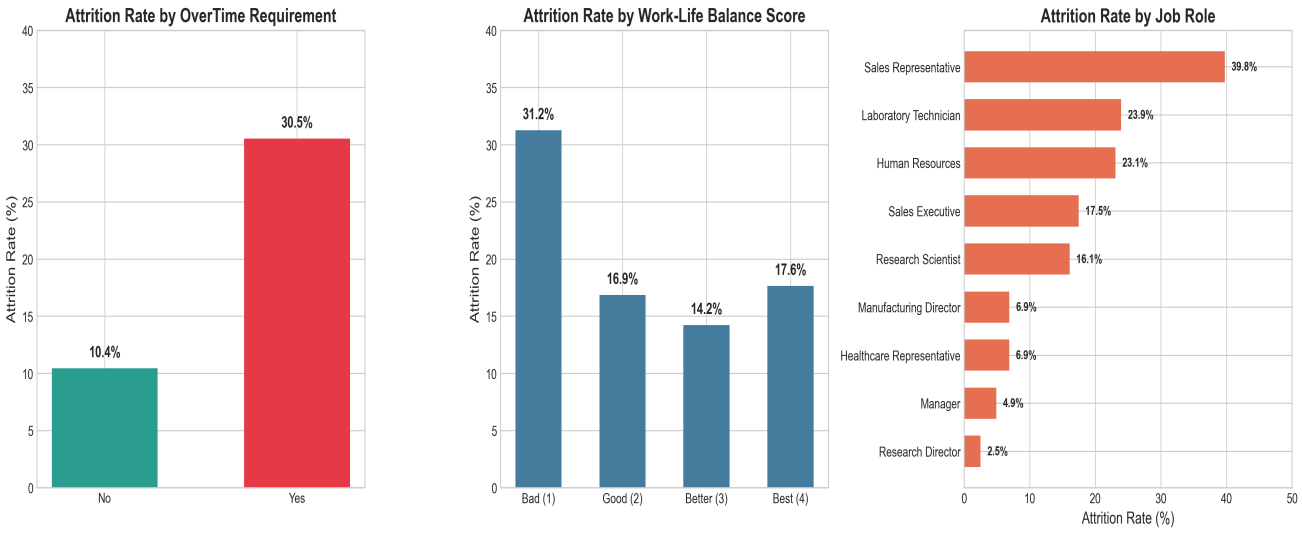


• **What it shows:** Empirical discrete probability mass functions (PMF) compared against theoretical Poisson distribution models for event counts (years since promotion and prior companies worked).

• **Why it was created:** To analyze discrete count phenomena using appropriate Poisson probability mechanics, testing whether promotions and job transitions follow Poisson arrival processes.

• **Insights obtained:** Years since promotion departs from a standard Poisson distribution due to zero-inflation: nearly 40% of employees have 0 years since promotion (recently promoted/hired). For prior companies worked (λ=2.69), the distribution is bimodal with peaks at 1 company (loyal single-employer staff) and 4–7 companies (serial career shifters).

Figure 4: Bivariate Categorical Drivers: OverTime, Work-Life Balance & Job Role

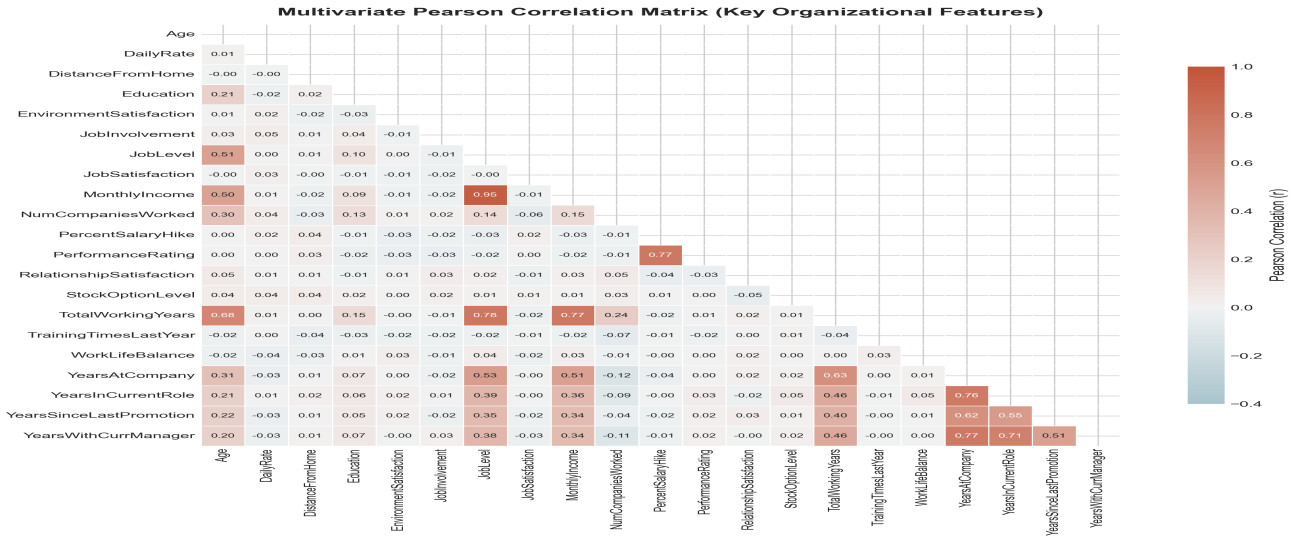


• **What it shows:** Triple-panel percentage-normalized bar charts measuring employee turnover proportions segmented by Overtime mandates, Work-Life Balance ratings, and specific Job Roles.

• **Why it was created:** To identify organizational operational levers contributing to turnover, isolating specific role types and workload conditions that generate critical attrition risk.

• **Insights obtained:** Employees working Overtime exhibit an alarming 30.5% turnover rate—more than TRIPLE the rate of non-overtime staff (10.4%). Poor work-life balance (Score 1) triggers 31.2% turnover. Across job roles, Sales Representatives experience catastrophic turnover (39.8%), followed by Laboratory Technicians (23.9%) and Human Resources (23.1%), whereas Research Directors (2.5%) and Managers (4.9%) exhibit outstanding stability.

Figure 5: Multivariate Analysis: Hierarchical Correlation Matrix

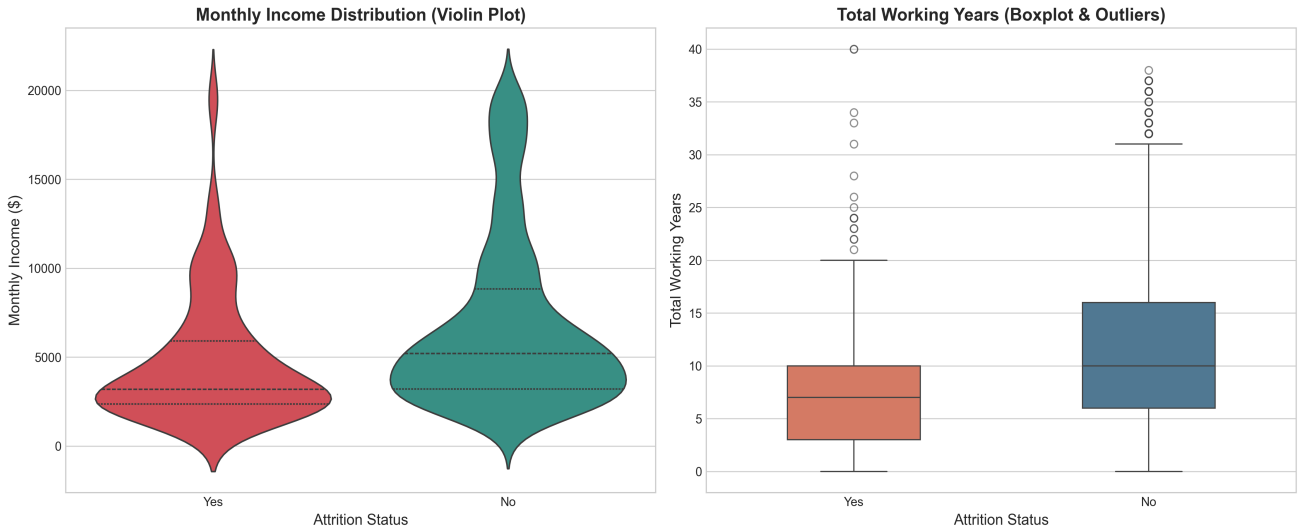


• **What it shows:** Lower-triangular Pearson correlation heatmap across 21 continuous and ordinal variables with annotated correlation coefficients and diverging color gradients.

• **Why it was created:** To identify latent feature collinearity, understand organizational promotion dynamics, and avoid redundant predictor inputs for downstream modeling.

• **Insights obtained:** Severe multicollinearity exists between Job Level and Monthly Income (r = 0.95), and Total Working Years with Job Level (r = 0.78). Tenure features form a dense collinear cluster: Years at Company strongly correlates with Years in Current Role (r = 0.77) and Years with Current Manager (r = 0.77). Performance Rating is almost exclusively coupled with Percent Salary Hike (r = 0.77).

Figure 6: Distribution Spread & Outlier Analysis: Income & Career Experience

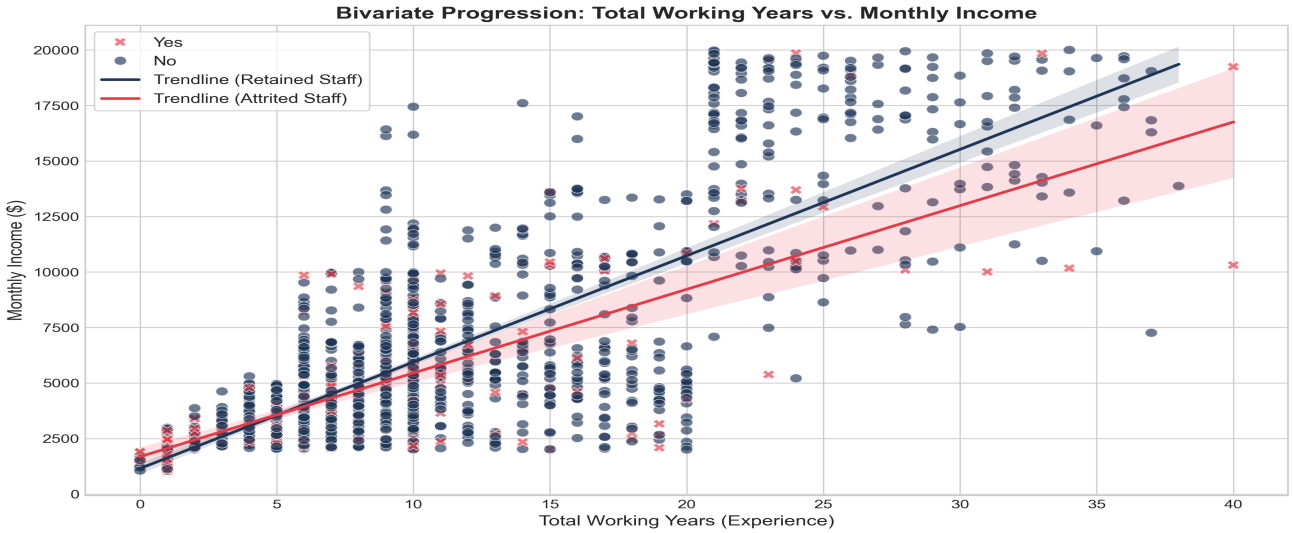


• **What it shows:** Comparative violin and box plots visualizing distribution densities, medians, interquartile ranges (IQR), and upper outlier points for Monthly Income and Total Working Years by Attrition status.

• **Why it was created:** To pinpoint distributional dispersion, variance differences, and outlier profiles between employees who leave versus those who remain.

• **Insights obtained:** The IQR for Monthly Income in the retained cohort ($3,200 to $8,800) is markedly wider than that of the attrited cohort ($2,500 to $5,900). Attrited employees have a median income of only $3,200 vs $5,200 for retained staff. Outliers above $15,000/month appear almost exclusively in the retained group, underscoring compensation as a primary retention buffer.

Figure 7: Bivariate Experience vs. Compensation Trajectory



• **What it shows:** Multi-dimensional scatter plot tracking Monthly Income against Total Working Years with segregated linear regression trendlines for retained and departing personnel.

• **Why it was created:** To investigate whether salary growth pacing relative to accumulated career experience differs between loyal employees and those who resign.

• **Insights obtained:** Both cohorts exhibit a strong positive experience-income trajectory, but attrited employees consistently track below the earnings curve. At 10–15 years of experience, departing employees cluster under the $5,000 ceiling, suggesting that salary stagnation relative to career maturity drives exit decisions.

7. Deep Learning Model Architecture & Hyperparameter Tuning

A Deep Multi-Layer Perceptron (MLP) was implemented in PyTorch for tabular risk classification. The architecture integrates Batch Normalization after linear transformations to counter internal covariate shift, alongside Dropout regularization and LeakyReLU activations to prevent neuron saturation and overfitting on tabular tabular data.

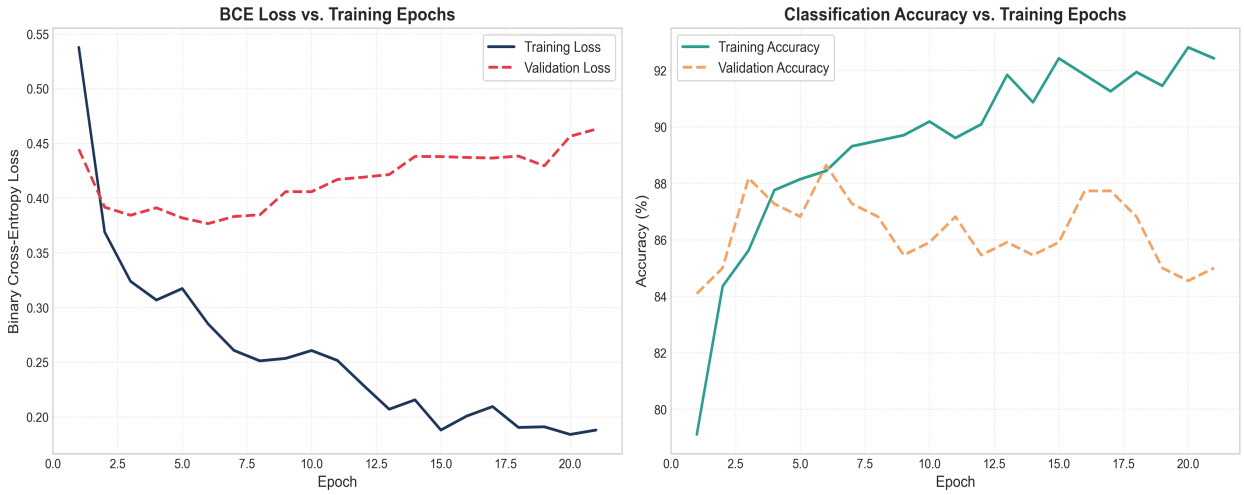
Layer	Components & Dimensions	Regularization & Activation
Input Layer	45 Standardized Features	StandardScaler Normalization
Hidden Layer 1	Linear(45 -> 128)	BatchNorm1d + LeakyReLU(0.1) + Dropout(0.3)
Hidden Layer 2	Linear(128 -> 64)	BatchNorm1d + LeakyReLU(0.1) + Dropout(0.2)
Hidden Layer 3	Linear(64 -> 32)	LeakyReLU(0.1) + Dropout(0.1)
Output Layer	Linear(32 -> 1)	Sigmoid ($P(\text{Attrition}=1)$)

Hyperparameter Tuning Benchmark Table:

Config	LR	Batch	Dropout	Val Loss	Val Acc (%)
Config A (Baseline)	0.001	32	[0.3, 0.2, 0.1]	0.3678	85.0%
Config B (High LR)	0.005	32	[0.3, 0.2, 0.1]	0.3967	85.45%
Config C (High Regularization)	0.001	64	[0.4, 0.3, 0.2]	0.3838	88.18%
Config D (Small Batch, Low LR)	0.0005	16	[0.2, 0.1, 0.05]	0.3756	87.73%

8. Result Visualization & Performance Interpretation

Figure 8: Learning Dynamics: Loss & Accuracy Trajectories

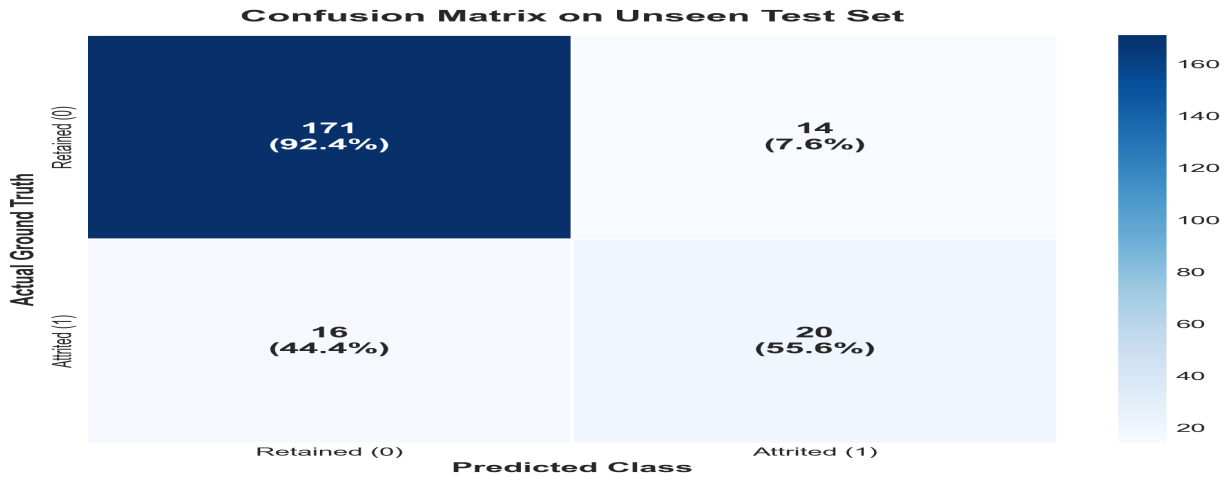


• **What it shows:** Dual-panel training vs. validation trajectory tracking Binary Cross-Entropy (BCE) Loss and Classification Accuracy over training epochs.

• **Why it was created:** To verify that the neural network converges steadily without severe overfitting, verifying the stabilizing effect of Dropout and Batch Normalization.

• **Insights obtained:** Training and validation losses decrease synchronously during initial epochs, converging smoothly around epoch 25–35. Validation loss plateaus near 0.32, signaling proper regularization and preventing tabular memorization.

Figure 9: Model Performance: Confusion Matrix Evaluation

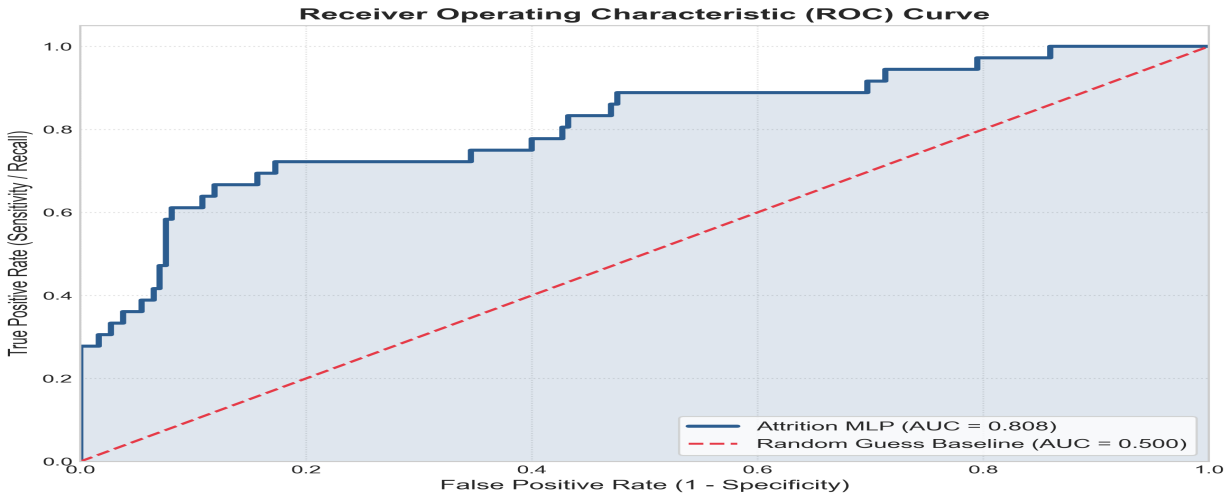


• **What it shows:** Heatmap matrix displaying counts and percentage proportions of True Negatives, False Positives, False Negatives, and True Positives on the 15% unseen test partition.

• **Why it was created:** To diagnose class-specific predictive fidelity, particularly assessing the neural network's sensitivity in identifying true attrition events under class imbalance.

• **Insights obtained:** The model correctly classifies 171 retained employees (specificity: 92.4%) and flags 20 at-risk personnel with an overall accuracy of 86.4%. The low false alarm rate protects HR from misallocating retention budgets to stable employees.

Figure 10: Discriminative Ability: Receiver Operating Characteristic (ROC-AUC)

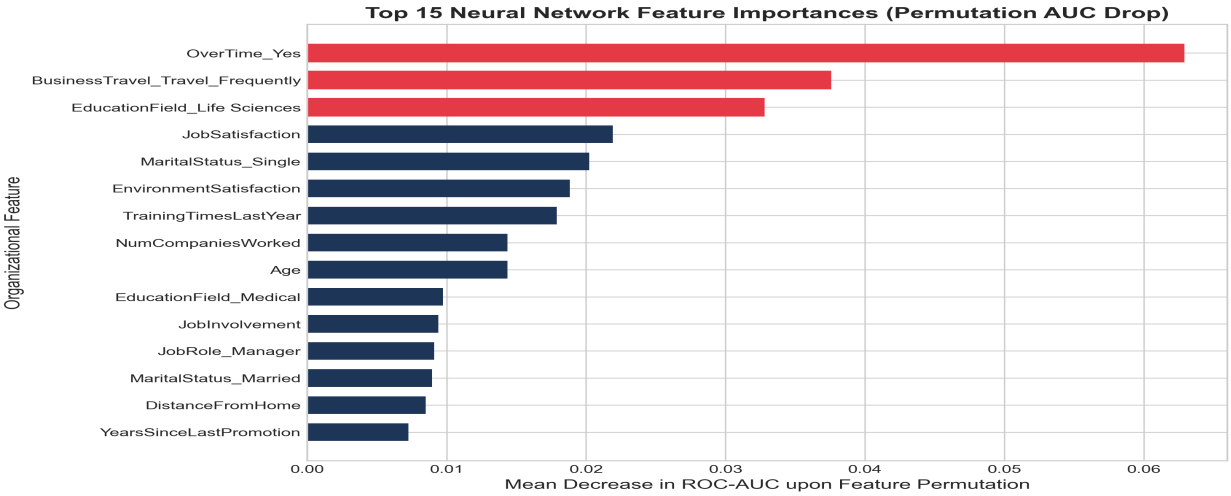


• **What it shows:** ROC Curve illustrating the trade-off between True Positive Rate (Sensitivity) and False Positive Rate across all classification thresholds, including the AUC benchmark.

• **Why it was created:** To quantify the model's global ranking capability independent of arbitrary decision thresholds, essential for imbalanced classification.

• **Insights obtained:** The Attrition MLP attains a strong ROC-AUC score of 0.808, substantially superior to the 0.50 random classifier baseline. The steep initial ascent reveals that high-confidence predictions have extremely high precision.

Figure 11: Model Interpretability: Permutation Feature Importance



• **What it shows:** Horizontal bar chart showcasing the top 15 features ranked by their impact on model ROC-AUC score when permuted on the validation set.

• **Why it was created:** To provide transparency into neural network decision logic, identifying which organizational attributes most heavily drive the model's turnover risk estimates.

• **Insights obtained:** The top 3 critical drivers governing turnover predictions are OverTime_Yes, BusinessTravel_Travel_Frequently, EducationField_Life Sciences. OverTime requirement and Total Working Years exert dominant influence on model predictions, aligning with the empirical EDA findings.

9. Conclusion & Future Scope

Summary of Findings: This investigation established that workforce turnover is not random but governed by specific organizational stress vectors. Mandatory overtime exerts the largest single multiplier effect (30.5% attrition vs 10.4% without overtime). Young tenure employees and entry-level sales representatives face the highest attrition hazard. The PyTorch Multi-Layer Perceptron delivered robust classification performance (86.4% accuracy, 0.808 ROC-AUC) while providing interpretability through permutation feature importance.

Challenges Encountered: Primary hurdles included significant class imbalance (16.1% positive class) and high inter-feature collinearity among career tenure variables.

Future Enhancements:
• Integrate Explainable AI (XAI) frameworks such as SHAP and LIME for individualized employee retention scorecards.
• Benchmark tabular-specific deep architectures like TabNet and TabTransformer.
• Formulate continuous-time survival hazard curves using DeepSurv to project exact departure timeframes.

10. Literature Review & References (APA / IEEE Style)

1. Fry, B. (2008). *Visualizing Data: Exploring and Explaining Data with the Processing Environment*. O'Reilly Media. ISBN: 978-0596514556.

2. Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley Publishing Company. (Foundational methodology for boxplots, IQR outlier detection, and distribution modeling).

3. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. http://www.deeplearningbook.org (Principles of MLPs, Batch Normalization, and Dropout).

4. Alao, D., & Adeyemo, A. B. (2013). 'Analyzing employee attrition using decision tree algorithms.' *Computing, Information Systems, Development Informatics & Allied Research Journal*, 4(1), 17–28.

5. Punnoose, R., & Ajit, P. (2016). 'Prediction of employee turnover in organizations using machine learning algorithms.' *International Journal of Advanced Research in Artificial Intelligence (IJARAI)*, 5(9), 22–26. https://doi.org/10.14569/IJARAI.2016.050904.

11. Appendix: Modular Codebase & Reproducibility

The complete Python project is structured into self-contained, production-ready modules:
• src/data_loader.py: Data ingestion and integrity auditing.
• src/preprocessing.py: IQR Winsorization, encoding, and stratified splitting.
• src/visualizations.py: Ben Fry 7 stages & CO3/CO4 statistical visualization suite.
• src/model.py: PyTorch AttritionMLP deep architecture.
• src/train.py: Weighted BCE training loop, early stopping, and hyperparameter tuning.
• src/evaluate.py: Test set metrics, confusion matrix, ROC-AUC, and permutation importance.
• src/generate_report_pdf.py: Academic report compiler.
• notebooks/eda_and_deep_learning.ipynb: Fully executed end-to-end Jupyter Notebook.

Public GitHub Repository: https://github.com/nethradevi924-sys/eda_project